

ROTATIONAL MOTION AND GRAVITATION

ROTATIONAL MOTION

When a body rotates about a fixed axis, it is said to perform a rotational motion and the axis is called the **axis of rotation**. In rotational motion every particle of the body moves in a circle and the centres of all these circles lie at the axis of rotation.

- The rotating blades of an electric fan and the motion of a top are examples of rotational motion.
- Rigid body is the body in which there is no displacement among the particles of the body on applying an external force.

Centre of Mass [CM]

A point within the body or outside the body at which the whole mass of the body is assumed to be concentrated.

- The centre of mass of sphere, cylinder and ring is at its geometric centre.
- The centre of mass may lie outside the body where there is no material as in case of ring, hollow sphere, hollow cylinder, etc.

Moment of Inertia

Moment of inertia of a body with respect to axis of rotation is the product of the mass (m) and the square of distance from the axis of rotation (R).

i.e. moment of inertia $I = mR^2$

- The SI unit of moment of inertia is kg-m^2 .
 - Moment of inertia of a ring about an axis passing through its centre and perpendicular to its plane is $I = MR^2$.
 - Moment of inertia of a diameter about its diameter is $I = \frac{1}{2}MR^2$.
 - The moment of inertia depends not only on the mass and size of the body but also upon the manner in which the mass is distributed around the axis of rotation.
- Practical applications of moment of inertia are given below
- The moment of inertia plays important role in our daily life.
 - In cycle, rikshaw, bullockcart, etc., the moment of inertia of the wheels is increased by concentrating most of the mass at the rim of the wheel and connecting the rim to the axle of the wheel through spokes.
 - It is due to the large moment of inertia of the wheels that when we stop cycling, the wheels of the cycle continue rotating for some time.

Radius of Gyration

Radius of gyration is defined as the distance from axis of rotation at which, if total mass of the body were supposed to be concentrated, the moment of inertia would be same.

$$I = MK^2$$

where, M = total mass of the body, K = radius of gyration

Torque or Moment of Force

The turning effect of a body is known as the moment of the force or torque.

- Moment of force or torque, $\tau = F \cdot d$, here d is the perpendicular distance from the axis of rotation.
- It is a vector quantity having SI unit N-m.
- A body of lesser weight can produce the same turning effect as the body of higher weight by adjusting their distance from the axis of rotation.
- Torque is equal to the product of moment of inertia and angular acceleration. i.e. $\tau = I \times \alpha$.

Practical applications of torque are given below

- The handle or the knob is fitted near the free edge of the door. So that, a heavy revolving door is opened or closed easily, when the force is applied at the edge of the door.
- A wrench with a long arm is used to unscrew a nut fitted tightly into a bolt and the handle of a screw driver is made wide.

ANGULAR MOMENTUM

- The angular momentum (J) of a body about an axis is equal to the product of the moment of inertia (I) of the body and its angular velocity (ω) about that axis. i.e. $J = I\omega$.
- If no external torque is acting upon a body rotating about an axis, then the angular momentum of the body remains constant. It is called conservation of angular momentum.

Examples related to conservation of angular momentum are given below

- When a diver jumps into water from a height, he does not keep his body straight but pulls in his arms and legs towards the centre of his body. On doing so, the moment of inertia of his body decreases. But, since the angular momentum remains constant, his angular velocity increases.
- A man with his arms outstretched and holding heavy dumb-bells in each hand, is standing at the centre of a rotating table. When the man pulls-in his arms, the speed of rotation of the table increases.
- The reason is that on pulling-in the arms, the distance of the dumb bells from the axis of rotation decreases and so the moment of inertia of the man decreases. Therefore by conservation of angular momentum, the angular velocity increases.
- The angular velocity of revolution of planet around the sun in elliptical orbit increases, when the planet comes closer to the sun.

- If a cat is dropped from a height, they almost always tend to land on their feet because they use the conservation of angular momentum to change their orientation.
- When a cat is in the air, no net external torque acts on it about its centre of mass, so the angular momentum about the cat's centre of mass cannot change.
- Stretching out its legs increases its rotational inertia and thus slows the cat's angular speed. The conservation of angular momentum allows the cat to rotate its body and slow its rate of rotation enough, so that it lands on its feet easily.

ROLLING MOTION

The rolling motion can be regarded as the combination of pure rotation and pure translation.

- The kinetic energy of a body rolling without slipping is the sum of kinetic energies of translational and rotational motion.

i.e. total KE of a rolling body

$$= \text{Rotational KE} + \text{Translational KE}$$

$$= \frac{1}{2} I \omega^2 + \frac{1}{2} m v_{cm}^2$$

- If a ring, a disc and a sphere, all of the same radius and mass roll down an inclined plane from the same height, the sphere will reach the bottom first and ring at last.

GRAVITATION

The phenomena of action of attractive forces between two bodies by virtue of their masses is known as gravitation.

Newton's Law of Gravitation

The force of gravitational attraction between two bodies is directly proportional to the product of their masses ($m_1 \cdot m_2$) and inversely proportional to the square of the distance (r)

between them i.e. $F = G \frac{m_1 m_2}{r^2}$

where, m_1 is the mass of first body, m_2 is the mass of second body and r is the distance between them.

- The value of G is $6.67 \times 10^{-11} \text{ N-m}^2/\text{kg}^2$.
- It is always attractive in nature, unlike electric and magnetic force which can be attractive or repulsive both.
- It acts along the line joining the centres of two interacting forces.
- The gravitational force between two masses is independent of the presence of any other object medium present between two masses.
- The predictions of solar and lunar eclipses on the basis of the Newton's law of gravitation come out in perfect agreement with actual observation.
- Tides are formed in oceans due to gravitational attraction between moon and ocean water.

Acceleration due to Gravity

It is the force by which the earth attracts a body towards its centre.

- The acceleration due to gravity is the rate of change of velocity of a body due to the gravitational force acting towards the earth. It is represented by $g = \frac{GM_e}{R_e^2}$

where, M_e is the mass of the earth and R_e is the radius of the earth.

- The value of g at the surface of the earth is 9.8 m/s^2 .
- Earth is surrounded by an atmosphere of gases due to gravity. The value of g on the moon is 1/6th of that of the earth.

Variation in the Value of (g)

- When we go above the surface of the earth, the acceleration due to gravity goes on decreasing.
- When we go below the surface of the earth, the acceleration due to gravity goes on decreasing and becomes zero at the centre of the earth.
- Decreasing the rotational motion of the earth, the value of g increases. When we go from the equator towards the poles, the value of g goes on increasing.
- If earth stops its rotation about its own axis, then at the equator, the value of g increases and consequently the weight of body lying there increases.
- The value of g is maximum on surface of the earth.

MASS AND WEIGHT

The mass of a body is the quantity of matter contained in it. It is a scalar quantity and its SI unit is kg. Mass does not change from place to place and remains constant.

The weight of the body is the force with which it is attracted towards the centre of the earth.

- Weight of the body $w = mg$
- Weight of a body is not constant, it changes from place to place. At the pole, the weight of a body will be maximum whereas at the equator it is minimum.
- Mass is measured by an ordinary equal arm balance, while weight is measured by a spring balance.

Weight of a Body at Moon

As mass and radius of moon is lesser than the earth, so the force of gravity at the moon is also less than the earth. Its value at the moon's surface is $g/6$.

Centre of Gravity

The centre of gravity of a body is that point, at which the whole weight of the body appears to act.

- It can be inside the material of the body or outside it.
- For regularly shaped body, the centre of gravity lies at its geometrical centre.

Weight of a Body in Lift

Some conditions are given below

- If lift is stationary or moving with uniform speed (either upward or downward), the apparent weight of a body is equal to its true weight.
- If lift is going up with acceleration, the apparent weight of a body is more than the true weight.
- If lift is going down with acceleration, the apparent weight of a body is less than the true weight.
- If the rod of the lift is broken, it falls freely. In this situation, the weight of a body in the lift becomes zero. This is the situation of weightlessness.
- While going down, if the acceleration of lift is more than acceleration due to gravity, then the body in the lift raise towards the ceiling of the lift.

SATELLITE

The heavenly body which revolves around the sun is called **planet** e.g. the earth, while the heavenly body which revolves round the planets is called **Satellite**. Moon is a natural satellite of the earth.

- Orbital speed of a satellite is independent of its mass and depends upon the radius of orbit.
i.e. $v_0 = \sqrt{gR}$
- The orbital speed of a satellite revolving near the surface of the earth is 7.9 km/s.
- Time period of revolution of satellite revolving near the surface of the earth is 1 hr 24 min (84 min).
- Period of revolution of a satellite depends upon the height of satellite from the surface of the earth.
- Every body inside the satellite is in a state of weightlessness.
- Period of revolution of a satellite is independent of its mass.
- If a satellite revolves in equatorial plane in the direction of the earth's rotation i.e. from West to East with a period of revolution equal to time period of rotation of the earth on its own axis i.e. 24 h, then the satellite will appear stationary relative to the earth. Such a satellite is called **geo-stationary satellite**. Such a satellite revolves around the earth at a height of 3600 km.
- INSAT 2B and INSAT 2C are geo-stationary satellites of India. It is used to reflect TV signals and telecast TV programs from part of the work to another.
- **Polar satellite** revolves around the earth in polar orbit at a height of 800 km. Time period of the satellites is 84 min. These are used for weather forecasting and mapping, etc.
- **Note** The earth rotates on its axis from West to East. This rotation makes the sun and the stars appear to be moving across the sky from East to West.

Energy of a Satellite

A satellite revolving around the earth has potential energy as well as kinetic energy. It has potential energy because it remains in the gravitational field of earth and kinetic energy because it is in motion.

Total energy of the satellite,

$$E = \text{Kinetic energy} + \text{Potential energy}$$

$$E = \frac{GM_e m}{2R_e} + \left(-\frac{GM_e m}{R_e} \right)$$

$$E = -\frac{GM_e m}{2R_e}$$

where, m is the mass of satellite.

ESCAPE VELOCITY

Escape velocity is that minimum velocity with which a body should be projected from the surface of the earth, so as it goes out of gravitational field of earth and never returns to the earth. Escape velocity at the surface is 11.2 km/s.

Escape velocity is given by

$$v_e = \sqrt{\frac{2GM}{R_e}}$$

$$v_e = \sqrt{2gR_e} \quad \left[\because g = \frac{GM}{R_e^2} \right]$$

- The value of the escape velocity of a body does not depend on its mass. Its value on the moon surface is 2.38 km/s. So, there is no atmosphere around the moon.
- Escape velocity is $\sqrt{2}$ times the orbital velocity.
- Satellites are launched with the escape velocity as needed.

KEPLER'S LAWS

Kepler's law of planetary motion are three scientific laws describing motion of planets around the sun.

- All planets move around the sun in elliptical orbits having the sun at one focus of the orbit.
- The areal speed of a planet around the sun is constant.
- The square of the period of revolution of any planet around the sun (T) is directly proportional to the cube of its mean distance from the sun (a) i.e.

$$T^2 \propto a^3.$$

Black Hole

A black hole is a region of spacetime exhibiting such strong gravitational effects that nothing not even particles and electromagnetic radiation such as light can escape from inside it. Event horizon is the boundary marking the limits of a black hole. Thus, nothing that enters a black hole can get out or can be observed from outside the event horizon.

PROPERTIES OF MATTER

ELASTICITY

Elasticity is the property of material of a body by virtue of which the body acquires its original shape and size after the removal of deforming force.

- Steel and ivory are more elastic than rubber and water is more elastic than air.
- The internal restoring force acting per unit area of cross-section of a deformed body is called **stress**. Its unit is N/m^2 or Pascal.
- The fractional change in configuration i.e. length, volume and shape is called **strain**.
- The ratio of stress to strain is a constant for the material and is called **modulus of elasticity**.

i.e.
$$E = \frac{\text{Stress}}{\text{Strain}}$$

It is also called **Hooke's law**.

- The maximum deforming force up to which a body retains its property of elasticity is called the **limit of elasticity**.
- Putty, paraffin wax are nearly perfectly plastic bodies.
- Iron, copper, silver, aluminium, etc., are examples of ductile materials. Glass, dry clay, etc., are examples of brittle materials.
- Rubber is one example of elastomers.

PRESSURE

Pressure is defined as, force acting normally on unit area of the surface.

$$\text{Pressure} = \frac{\text{Force}}{\text{Area}}$$

- Its unit is N/m^2 also called Pascal. It is a scalar quantity.
- The pressure exerted by liquid at depth h below the surface of liquid is given as,

$$p = h \rho g$$
, where ρ is density of liquid.
- It is measured by manometer.
- Pressure at a point in a static liquid has same value in all directions.
- Pressure at a point in a liquid is proportional to the density of the liquid and depth of the point from the free surface.
- Boiling point of all substances increases with the increase in pressure.

Atmospheric Pressure

The pressure exerted by the atmosphere is called atmospheric pressure.

- Atmospheric pressure = 1.01 bar

$$= 1.01 \times 10^5 \text{ N/m}^2$$

$$= 760 \text{ torr}$$
- Atmospheric pressure is measured by barometer.
- Atmospheric pressure decreases with altitude (height from the earth's surface). This is why
 - It is difficult to cook on the mountain.
 - The fountain pen of a passenger leaks in aeroplane at height.
- The slow rise in the barometric reading is the indication of clear weather.
- Sudden fall in barometric reading is the indication of storm.
- Slow fall in barometric reading is the indication of rain.
- **Note** Building and dams are laid on a larger area of ground so that, the height of the building or dam produces less pressure on ground.

PASCAL'S LAW

Pascal's law states that 'Pressure in a fluid in equilibrium is the same everywhere, if the effect of gravity can be neglected.'

- Hydraulic lift, hydraulic press, hydraulic brakes work on the basis of Pascal's law.
- Hydraulic lift is used to lift heavy loads.

ARCHIMEDES' PRINCIPLE

When a solid body is immersed wholly or partially in a liquid (in general, in a fluid), then there is some apparent loss in its weight. This loss in weight is equal to the weight of the liquid displaced by the body.

Applications of Archimedes' principle are given below

- Archimedes' principle is used in determining the relative density of a substance.
- Archimedes' principle is used in designing ships and submarines.
- The hydrometers used for determining the density of liquids are based on Archimedes' principle.
- The lactometers used for determining the purity of milk are based on Archimedes' principle.

DENSITY

Density is defined as mass per unit volume.

- Density, $\rho = \frac{\text{Mass}}{\text{Volume}} = \frac{M}{V}$. Its SI unit is kg/m^3 .
- **Relative density** of substance is defined as the ratio of the density of that substance to the density of water at 4°C i.e.

$$\text{Relative density} = \frac{\text{Density of material}}{\text{Density of water at } 4^\circ\text{C}}$$

- Relative density has no unit. It is a scalar quantity.
- The density of water is maximum at temperature 4°C .
- Relative density of liquid or fluid is measured by **hydrometer**.
- The density of sea water is more than that of normal water. This explains why it is easier to swim in sea water.
- If ice floating in water in a vessel melts, the level of water in the vessel does not change.
- Ice floats on water surface as its density (0.92 g/cm^3) is lesser than the density of water (1 g/cm^3).
- Iron will sink in mercury because density of mercury is more than the iron.

BUOYANT FORCE

When a body is immersed partially or wholly in a liquid, a force acts on the body by the liquid in the upward direction. This force is called buoyant force or force of buoyancy or upthrust.

- It is equal to the weight of liquid displaced by the body and acts at the centre of gravity of displaced liquid.
- The buoyant force exerted by a liquid depends on the volume of the solid object immersed in the liquid.
- The buoyant force exerted by a liquid depends on the density of the liquid in which the object is immersed.

Law of Floatation

When body floats in neutral equilibrium, the weight of the body is equal to the weight of displaced liquid.

- The centre of gravity of the body and centre of gravity of the displaced liquid should be in one vertical line.
- Three cases are possible when the body is immersed in the fluid.
 - (i) If the weight of the body is greater than the upward force or upthrust by the fluid acting upwards, then the body sinks.
 - (ii) If the weight of the body is equal to the upthrust or the weight of the body is just balanced by upthrust, then the body floats fully immersed.
 - (iii) If the weight of the body is less than the upward force, then the body floats partially immersed. The centre of gravity of a body is that point at which the whole weight of the body appears to act.

Fluid Dynamics

- (i) **Streamline Flow or Steady Flow** In this flow velocity at every point in the fluid will remain constant.
- (ii) **Turbulent Flow** In this flow, the fluid does not maintain constancy of the velocity.
- (iii) **Critical Velocity** The velocity above which flow will become turbulent.

PRINCIPLE OF CONTINUITY

It states that, when an incompressible and non-viscous liquid flows in streamlined motion through a tube of non-uniform cross-section, the product of the area-section and the velocity of flow is same at every point in the tube. For a tube of flow, between two points having area of cross-section A_1 and A_2 and velocities V_1 and V_2 .

Then,

$$AV = \text{Constant}$$

$$A_1V_1 = A_2V_2$$

According to the equation of continuity, the speed of fluid flow becomes faster in the narrower pipe.

BERNOULLI'S THEOREM

When an incompressible and non-viscous liquid (or gas) flows in streamlined motion from one place to another, then at every point of its path the total energy per unit volume (Pressure energy + Kinetic energy + Potential energy) is constant.

$$P + \frac{1}{2}\rho v^2 + \rho gh = \text{constant}$$

where, ρ is density of fluid. This is the principle of conservation of energy for the non-viscous fluids i.e. there is no viscous force acting between the layers of the fluid.

Venturimeter, Pitot tube, Bunsen's burner, atomizer, filter pump and magnus effect are based upon the Bernoulli's theorem.

- **Note** Bernoulli's theorem is applicable only to streamline flow of a fluid. It is not valid for non-steady or turbulent flow.

SURFACE TENSION

Surface tension of a liquid is measured by the normal force acting per unit length on either side of an imaginary line drawn on the free surface of liquid and tangential to the free surface.

- If a force F acts on an imaginary line of length l , then surface tension $T = \frac{F}{l}$. Its SI unit is Newton/metre.
- Force of attraction applied between molecules of same substance is called **cohesive force**, while attractive force between molecules of different substances is called **adhesive force**.